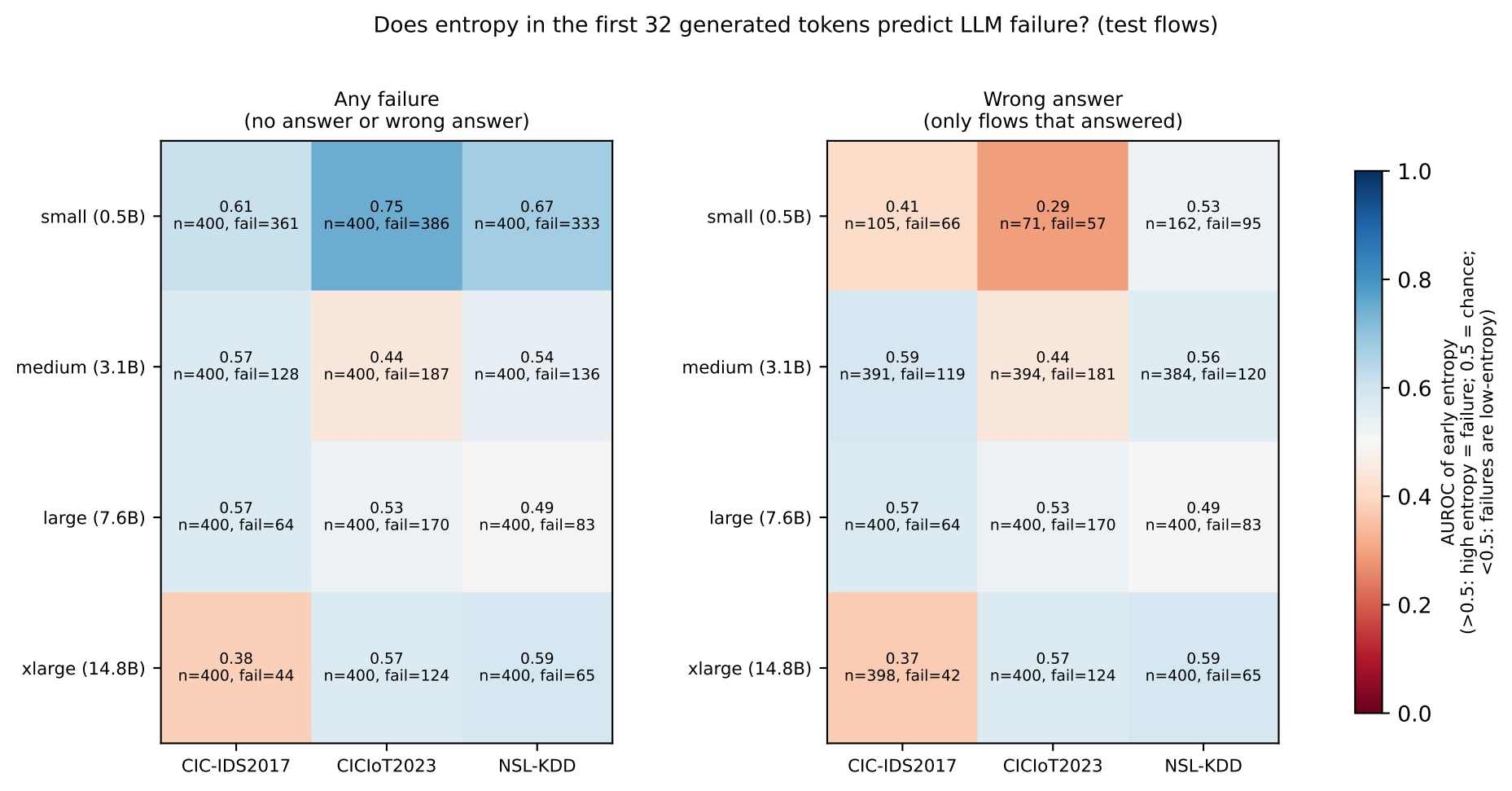
Does entropy in the first 32 generated tokens predict LLM failure? (test flows)
Any failure
(no answer or wrong answer)
Wrong answer
(only flows that answered)
small (0.5B)
medium (3.1B)
large (7.6B)
xlarge (14.8B)
0.61
n=400, fail=361
0.75
n=400, fail=386
0.67
n=400, fail=333
0.57
n=400, fail=128
0.44
n=400, fail=187
0.54
n=400, fail=136
0.57
n=400, fail=64
0.53
n=400, fail=170
0.49
n=400, fail=83
0.38
n=400, fail=44
0.57
n=400, fail=124
0.59
n=400, fail=65
0.41
n=105, fail=66
0.29
n=71, fail=57
0.53
n=162, fail=95
0.59
n=391, fail=119
0.44
n=394, fail=181
0.56
n=384, fail=120
0.57
n=400, fail=64
0.53
n=400, fail=170
0.49
n=400, fail=83
0.37
n=398, fail=42
0.57
n=400, fail=124
0.59
n=400, fail=65
CIC-IDS2017
CICIoT2023
NSL-KDD
AUROC of early entropy
(>0.5: high entropy = failure; 0.5 = chance;
<0.5: failures are low-entropy)
1.0
0.8
0.6
0.4
0.2
0.0